Vaghesan Sundaram

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Education

University of Maryland, College Park — College Park, MD
B.S. Computer Science (Cybersecurity), B.S. Mathematics, ACES Cybersecurity Minor — *Expected December 2027*
Relevant Coursework: Data Structures, Algorithms, Computer Systems, Discrete Structures, Linear Algebra, Statistics, Cryptography, Network Security, Penetration Testing, Technical Writing, Programming Languages, Computer Networks, Compilers, Combinatorics and Graph Theory, Number Theory

Middlesex College — Edison, NJ
A.S. Mathematics, completed during high school — *June 2024*

Experience

Incoming Technology Intern (TIP) — Summer 2027
Capital One — *McLean, VA*

Cybersecurity Intern — June 2026-Present
Redmind Technologies — *Remote*
- Identified and documented 7 reproducible findings across authentication, data exposure, configuration, and dependency risk by combining Semgrep SAST with manual secure code review of a public-facing client application.
- Delivered a developer-facing report and slide briefing that validated evidence, prioritized all 7 findings by severity, and mapped each issue to actionable remediation guidance for developer implementation and client review.
- Developing a PyTorch pipeline with MuJoCo, robosuite, robomimic, and LeRobot to measure training-data and reward-label poisoning across behavior cloning, transformer, VLA, and offline RL policies.

Information Technology Intern — June 2025-August 2025
New Jersey Courts — *Toms River, NJ*
- Installed courtroom recording hardware to capture legal proceedings and support court recordkeeping.
- Improved IT asset inventory accuracy across 5 departments (Finance, IT, Civil, Probation, and Operations) by cataloging devices and reconciling physical equipment with tracking records for asset control.

Projects

Agent Least Privilege | *Python, MCP, Pydantic, asyncio, pytest, mypy, Ruff, GitHub Actions* — 07-26 | GitHub
- Built a Python MCP proxy enforcing launch-time policies for filesystem paths, exact arguments, and call quotas.
- Serialized call quotas, recorded redacted audit events, failed closed on errors, and tracked unknown outcomes.
- Blocked 12/12 unauthorized proposals; completed 10/12 attack tasks and 24/24 controls in simulation.

CTI Claim Provenance | *Python, CISA KEV/CSAF, NVD, CVE, MITRE ATT&CK, Pydantic* — 09-26 | GitHub
- Built a 64-question, 24-group point-in-time CTI benchmark; constrained output raised evidence binding 83.3% to 86.5% but lowered exact accuracy 33.3% to 26.0%; a 144-response follow-up found no schema benefit.
- Tracked CISA KEV/CSAF, NVD, CVE, MITRE ATT&CK, and vendor source provenance with SHA-256 hashes.
- Enforced point-in-time cutoff and source-authority rules with evidence spans, abstention, and human review.
- Automated typed-schema validation, split isolation, abstention, secret scanning, and privacy-safe aggregate checks.

Prophet Prediction Market | *Python, WebSockets, sentence embeddings, clustering* — 01-26 | GitHub
- Contributed to an autonomous prediction-market platform integrating xAI Grok event detection, OpenRouter trading agents, and Solana smart contracts for on-chain market creation and trading at HackPrinceton.
- Built a 4-route Flask pipeline processing up to 40 X posts/min by weighting 7 signals and clustering 10+ posts.

Keystroke Authentication System | *Python, Flask, SQLite, JWT, bcrypt, CSRF, CORS* — 03-26 | GitHub
- Secured 2 Flask services using 3-sample Pearson/MSE keystroke profiles, 20-character bcrypt fallback, 5-attempt/15-minute rate limiting, 60-second single-use codes, 1-hour JWTs, and atomic profile re-enrollment.
- Hardened login via CSRF, CORS allowlisting, JWT binding, callback validation, and server-to-server exchange.

Technical Skills

Certification: CompTIA Security+ (SY0-701), March 2026-March 2029
Languages: Python, Java, C++, C, Assembly, Rust
Security: OSINT, Maltego, Shodan, Recon-ng, Maigret, Burp Suite, Nmap, Wireshark, Ghidra, vulnerability triage, TCP/IP, DNS, DHCP, PCAP analysis, Kali Linux, Hashcat, John the Ripper, privilege escalation, exploit development
Tools: Git, GitHub Actions, Docker, Linux, PowerShell, Flask, Pydantic, pytest, Hypothesis, mypy, REST APIs, GDB
Data & ML: PyTorch, MuJoCo, robosuite, robomimic, LeRobot, SQL, SQLite, NumPy, SciPy, scikit-learn